MILITARY SPECIFICATION SHEET

Ⓑ ELECTRON TUBE, TRANSMITTING

TYPE 1624

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Beam power amplifier

Outline	---	16-2 (EIA)
Base	---	A5-11, low-loss phenolic
Envelope	---	ST-16
Ⓑ Cap	---	C1-1
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	Cap
Element	---	+f	g2	g1	nc	-f, g3	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ic1	Pg2	Pp	Pi	Modulation	Alt
Unit:	V	Vdc	Vdc	Vdc	mAdc	mAdc	W	W	W		ft
Maximum:											
AF Class AB2:	2.5 ±5%	600	--	300	90	---	3.5	25	54.0	---	10,000
Telep:	2.5 ±5%	600	-200	300	75	---	2.5	25	37.5	Grid	10,000
Telep:	2.5 ±5%	500	-200	300	75	5	2.5	16.5	37.5	Anode	10,000
Teleg:	2.5 ±5%	600	-200	300	90	5	3.5	25	54.0	---	10,000
TEST CONDITIONS:	2.5 Vdc	600	-16.5	300	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	1, 2	Ec1/Ib = 42 mAdc	0.65	II	Ic1	0	-5.0	µAdc
1256	Electrode current (anode)	-		0.65	II	Ib	28	52	mAdc
1256	Electrode current (screen)	-		0.65	II	Ic2	0	7.5	mAdc
1231	Emission	1	Eb = Ec1 = Ec2 = 50 Vdc	0.65	II	Is	200	---	mAdc
(B) ---	Short and discontinuity detection	5		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1306	Transconductance	-	Ec1/Ib = 48 mAdc; Eb = 300 Vdc	---	---	Sm	3,000	5,000	µmhos
1301	Filament current	-		---	---	If	1.85	2.15	A
1236	Power oscillation (1)	3	Ec2 = 200 to 300 Vdc; F = 15 MHz; Ib = 90 mAdc; Rg1 = 10,000 ohms; Ic1 = 5 to 7 mAdc	---	---	Po Pg2	27 ---	--- 3.5	W W
1331	Direct-interelectrode capacitance	-	Shield No. 312 Without shield Without shield	---	---	Cgp Cin Cout	--- 8.0 5.5	0.25 14.0 9.5	pF pF pF
1031	Low-frequency vibration	-	Eb = 250 Vdc; Ec2 = 100 Vdc; Ec1 = -5 Vdc; Rp = 2,000 ohms	---	---	Ep	---	1,000	mVac
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1111	Base pin solder depth	-		---	---	---	---	---	---
(B) 1211	Insulation of electrodes	6		4.0	S3	R	10	---	Meg
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
(B) ---	Life test	6	Group C; Ec1 = -17.8 Vdc	---	---	---	---	---	---
---	Life-test end points: (500 hours)								
1231	Emission or	-		---	---	Is	100	---	mAdc
1306	Transconductance	-		---	---	Sm	2,600	---	µmhos
(B) 1236	Power oscillation (2)	4	Power oscillation (1); F = 60 MHz	---	---	Po	23	---	W

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. An approved alternate test is, Ec1 = -16.5 Vdc with Ic1 = -40 µAdc maximum.
3. If failure occurs, 25 more tubes shall be tested, with no additional failures permitted; otherwise this test shall become a quality conformance inspection, part 1, test

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NOTES: -Continued.

(B) 4. This test shall be performed during the initial production and once each succeeding 12 calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of 3 tubes with an acceptance number of zero, and a second sample of 3 tubes with a combined acceptance number of 2. In the event of failure, the test will be made as a part of quality conformance inspection (QCI), part 2, code level D, with an AQL of 6.5.

(B) 5. A hexaphase short and discontinuity system may be used as an alternate test approach.

(B) 6. Use a 500 volt megger, or equivalent.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 80
DSA - ES

User activities:
Army - ME
Navy - AS, OS, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2900)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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OMB Approval No. 22-R255

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1807

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